



GRAPHICS CARD

A properly configured graphics card can spell the difference between you being the fragger or the frag-ee!

Arguably one of the most tweakable components in your computer, this is one device where any kind of bottleneck or chokepoint becomes instantly noticeable—if your graphics card dropped frames and slowed down gameplay during a *Quake III* deathmatch, you'd know about it faster than you could say fragbait!

For most of us, the most economical choice of graphics subsystem is that which is integrated in the motherboard itself. While purists would smirk at this approach, there is nothing to beat integrated graphics if you're on a budget. However, if you have a system with an integrated

graphics solution such as an Intel 815 motherboard or even the newer 845G chipset based board, your F1 Ferrari in that latest racing game may well respond like a Morris Minor! This is because your integrated graphics chipset doesn't have the processing horsepower to cope with the demands of today's games and applications. Onboard graphics are probably the least suited to intensive gaming and graphically demanding applications, as they just do not have the necessary power to deliver when it comes to pushing millions of pixels per second or swapping megabytes of environmental textures as seen in most new 3D games. At the very least, if your system does have integrated graphics, an AGP slot would give you the leeway to expand to a faster solution at a later date.

3D games today use enormous amounts of texture, vertex and model information, all of which require large amounts of memory to run. Since the

magic of 3D games is because they are 'realtime', all this texture swapping and polygon pushing has to happen at least at 30 fps. This calls for an extremely fast Video RAM. The most commonly used type of RAM on today's graphics cards is DDR RAM and it comes with 32 MB,

64 MB and in some cases even 128 MB of dedicated video memory on the card itself. An integrated graphics system does not

come with inbuilt memory and borrows it from your main system. This sharing of memory degrades the performance of a system using integrated graphics, as the RAM on your motherboard is significantly slower than RAM integrated

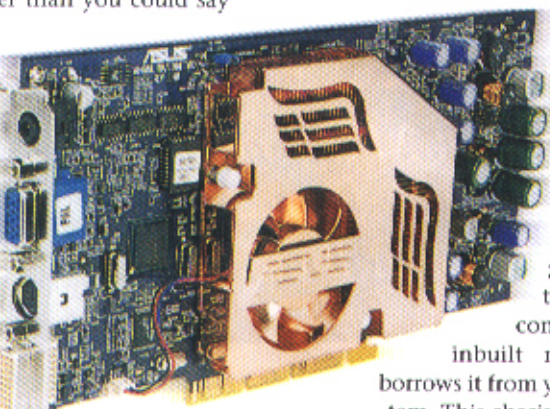
with the new-generation graphics cards.

As with processors, the whizbang features of the latest graphics cards need to be supported by the games you play. Therefore, while terms like TNL (Transform and Lighting), EMBM (Environmental Bump Mapping), Programmable Pixel and Vertex Shaders represent path-breaking technologies that promise greater levels of realism, you also need games that actually make use of these technologies. If you find that the mud on the 4x4 truck is not as realistic as you thought it would be with your new GeForce 4 card, it could mean that the game itself is not aware of the new multi-texture processing capabilities of the card.

Your graphics subsystem also needs to adhere to APIs (Application Programming Interfaces) such as DirectX or OpenGL. All games are built around either of these APIs—if your graphics card is not able to even launch the game, it could mean that the drivers do not

support these APIs. Newer driver updates for the graphics card will fix this. There are also numerous tweaks that can help extract a little extra performance out of your graphics card.

Lastly, even though your motherboard features an AGP slot, it does not necessarily mean that it is equipped to run tomorrow's graphics cards. AGP slots also have different standards that dictate how fast the AGP interface operates. The most popular interface right now is AGP 4x, which is practically standard on all motherboards and sports a bandwidth of 1 GBps. Future iterations such as AGP 8x will scale to higher speeds to provide room for the increasing bandwidth requirements of next-generation graphics cards.



Even with the formidable GeForce 4 graphics card, there are potential bottlenecks that can throttle your fps

Tips & Tweaks

- Always check for updates for your graphics card's drivers from the manufacturer's Web site. This gives you added functionality, improved performance and additional features, and quells bugs.
- Disable the 'Sync to Retrace' setting from your graphics card's driver if you are getting sub-par frame rates.
- If you are using a slower graphics card, try disabling environmental effects such as fog and smoke, and reduce the level of detail for the textures, characters and geometry to get those extra frames per second, albeit at the cost of graphical realism.
- Most graphics cards' drivers have a setting for enabling a function called anti-aliasing. While this setting reduces the effect of jaggies on screen (especially at lower resolutions), enabling this feature will lower your frame rates. Experiment with this setting and enable it only if you still have playable game speeds.
- On systems with integrated graphics chipsets, you can assign a greater amount of system memory as graphics memory—do this by increasing the AGP Aperture Size in your CMOS.